

Analysis–2, Test 3/Retake test from Part 2, 24th of May 2022.

- 1. (5+5 points)** Evaluate the following integrals :

$$\text{a) } \int \frac{\tan x}{\cos^3 x} dx \quad (x \in (-\pi/2; \pi/2)); \quad \text{b) } \int x^3 \cdot \ln x dx \quad (x > 0).$$

- 2. (8 points)** Evaluate the area of the closed and bounded region enclosed by the following three curves :

$$y = \sqrt{x}; \quad y = \sqrt{2x-1} \quad \text{and} \quad y = 0.$$

- 3. (8 points)** Find the volume of the rotation solid obtained by rotating the graph of the following function around axes x :

$$f(x) = \frac{\sqrt{7}}{\sqrt{10-x^2-3x}} \quad (x \in [0, 1]).$$

- 4. (6 points)** Find the local extrema places and values for the function :

$$f(x, y) := x^2 - y^3 + 2xy - 6x - y + 1 \quad ((x, y) \in \mathbb{R}^2).$$

- 5. (8 points)** Evaluate the following double integral :

$$\int \int_D e^{2x+y} dx dy,$$

where D is the closed triangle with vertices at points $A(0, 0); B(0, 2); C(1; 1)$.